

Radchaneeporn Changpun

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

PROFESSIONAL SUMMARY

AI and Data Science Professional who designs agentic systems from experiment through to deployment. Research background in multi-agent LLM frameworks built with clinical domain experts; independently deployed a multi-turn LLM chat service (FastAPI, PostgreSQL, Docker). Currently client-facing, combining data science, data engineering and business analytics to turn client data into decisions. M.Sc. Computer Science (GPA 3.94/4.00), Faculty of Engineering, Chulalongkorn University; published research on an agentic LLM framework for mental health support; AWS Certified AI Practitioner.

EDUCATION

M.Sc. in Computer Science, Chulalongkorn University

Aug 2023 – Dec 2025

Faculty of Engineering (GPA 3.94 /4.00)

Relevant coursework: Artificial Intelligence, NLP, Pattern Recognition, Data Science & Data Engineering Tools, Data Science Architecture, Computer Security, Research Methods in CS

Publication: Changpun, R. et al., "Agentic Stage-Based LLM Framework for Multi-Turn Mental Health Support Conversations in Thai," ISAI-NLP 2025, IEEE. [\[Paper\]](#)

B.Sc. in Chemical Engineering, Chulalongkorn University

Aug 2017 – May 2021

Faculty of Science

Publication: Nimmanterdwong, P.; Changpun, R. et al., "Applied Artificial Neural Network for Hydrogen Sulfide Solubility in Natural Gas Purification," ACS Omega, 2021. [\[Paper\]](#)

WORKING EXPERIENCE

PulseMetrics Pte Ltd

Oct 2025 – Present

Senior Associate, Data Science

- Work with the client's CRM team across two business units (Retail and Automotive Fast Fit), framing business questions and turning EDA, customer segmentation and ad-hoc analysis into clear, actionable insights for data-driven decisions
- Engineer high-performance, multi-threaded preprocessing pipelines for large-scale telematics data, cutting processing time by 40% (a 1.67x speedup)

AIMET, Center of Excellence in Digital and AI for Mental Health

Jul 2024 – Aug 2025

Researcher

- Designed and led the build of an end-to-end multi-agent LLM system for Thai mental health support, with human-in-the-loop controls and responsible AI practices for a sensitive, regulated domain; collaborated with a cross-functional team of researchers, developers and domain experts
- Owned the research lifecycle, running experiments across multiple approaches and building the AI evaluation and feedback loops with psychology experts; the system reached a 79% positive user-reaction rate and improved 7 of 8 support metrics across general and mental-health-specific evaluation rubrics

Wood PLC (Foster Wheeler)

Jan 2022 – Jan 2023

Chemical Process Engineer

- Designed and optimized chemical and energy processes for Thai oil and gas refinery clients at an international engineering consultancy

SELECTED PROJECTS

Production Multi-Turn LLM Chatbot: Built & Deployed End to End [\[GitHub\]](#)

- Designed, built and deployed a three-tier LLM chat service (Next.js, FastAPI, PostgreSQL) in which only the frontend is publicly exposed, containerized with Docker Compose for reproducible deployment
- Engineered multi-turn context management (sliding window plus rolling summarization) to keep conversations of any length within a fixed per-request token budget, and enforced a 20 requests/hour limit and a 50,000 output-token daily cap before every model call, keeping worst-case API spend bounded

LLM-Based Thai Personal Income Tax Q&A Chatbot [\[GitHub\]](#)

- Ran an empirical architecture evaluation across four LLM patterns (Agentic RAG, Naive RAG, Long-Context LLM, Vanilla Inference) to select the best approach for a regulated domain
- Assessed trade-offs with NLP metrics (BLEU, ROUGE-L, BERTScore) and LLM-as-a-judge, demonstrating evidence-based architecture decisions
- Built the data pipeline with OCR and web scraping; deployed an interactive UI with Gradio

RAG Pipeline with Open-Source LLMs (Llama 2-13B) [\[GitHub\]](#)

- Architected a RAG pipeline grounding an open-source LLM (Llama 2-13B) in a research literature corpus from Scopus, using Pinecone for embedding-based semantic retrieval
- Defined the retrieval pattern end to end (prompt orchestration, vector search, grounding) with LangChain to reduce hallucinations

Multi-Label Text Classification with RoBERTa [\[GitHub\]](#)

- Fine-tuned RoBERTa on Scopus data for a 40.3% Macro-F1 improvement (0.6687 vs 0.1894 baseline) via transfer learning and a custom architecture, using PyTorch and Hugging Face

Radchaneeporn Lab: Technical Content & Documentation [\[YouTube\]](#) [\[GitHub\]](#)

- Publish tutorials on modern LLM frameworks and document strategies for LLM cost optimization and system reliability

SKILLS

- **LLM & Agents:** RAG, Agentic RAG, multi-agent orchestration, fine-tuning (LoRA, QLoRA); LangGraph, LangChain, LlamaIndex, Hugging Face, Anthropic API
- **Evaluation:** Experiment design, LLM-as-a-judge, BLEU/ROUGE-L/BERTScore; Weights & Biases
- **Production:** FastAPI, Docker, PostgreSQL, REST APIs, rate limiting and cost control; AWS, Railway, Git, Linux
- **ML & Data:** Transformers, classification, clustering, feature engineering, data pipelines; Python, SQL, PyTorch, scikit-learn, Pandas

CERTIFICATION

AWS Certified AI Practitioner, Amazon Web Services [\[Credentials\]](#)

Mar 2026